
Appendix B

Laboratory Test Results

Material Test Report



[Signature]

Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

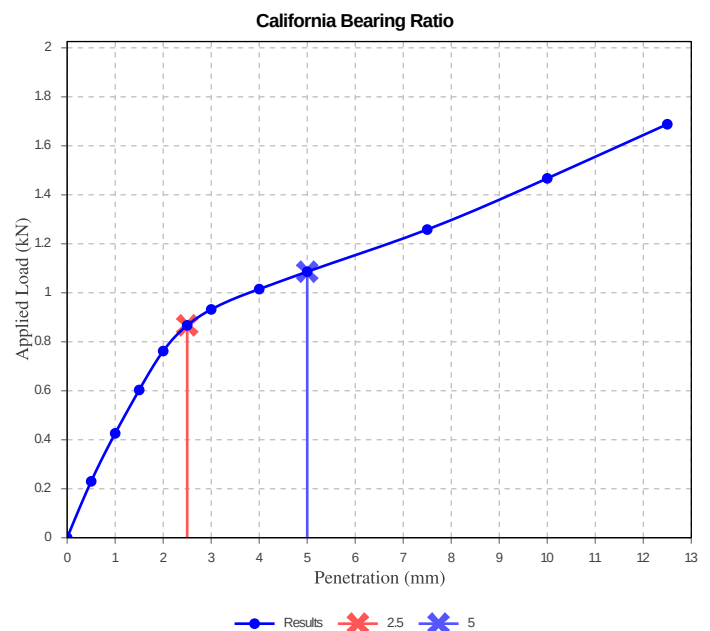
Report Number: 94417.00-1
Issue Number: 1
Date Issued: 12/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3498
Sample Number: GU-3498A
Date Sampled: 28/01/2020
Dates Tested: 28/01/2020 - 07/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Sample Location: Bore 1, Depth: 0.6-0.8m
Material: Sandy Clay

California Bearing Ratio (AS 1289 6.1.1 & 2.1.1)		Min	Max
CBR taken at	2.5 mm		
CBR %	7		
Method of Compactive Effort	Modified		
Method used to Determine MDD	AS 1289 5.2.1 & 2.1.1		
Method used to Determine Plasticity	Visual Assessment		
Maximum Dry Density (t/m ³)	1.95		
Optimum Moisture Content (%)	12.0		
Laboratory Density Ratio (%)	99.5		
Laboratory Moisture Ratio (%)	101.0		
Dry Density after Soaking (t/m ³)	1.87		
Field Moisture Content (%)	11.8		
Moisture Content at Placement (%)	12.3		
Moisture Content Top 30mm (%)	22.1		
Moisture Content Rest of Sample (%)	15.8		
Mass Surcharge (kg)	4.5		
Soaking Period (days)	4		
Curing Hours	71		
Swell (%)	4.0		
Oversize Material (mm)	19		
Oversize Material Included	Excluded		
Oversize Material (%)	0		

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	31		
Plastic Limit (%)	13		
Plasticity Index (%)	18		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	9.0		
Cracking Crumbling Curling	Cracking		

Moisture Content (AS 1289 2.1.1)	
Moisture Content (%)	11.8



Material Test Report



[Signature]

Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Report Number: 94417.00-1
Issue Number: 1
Date Issued: 12/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3498
Sample Number: GU-3498B
Date Sampled: 28/01/2020
Dates Tested: 28/01/2020 - 07/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Sample Location: Bore 4, Depth: 0.7-0.9m
Material: Sandy Silty Clay

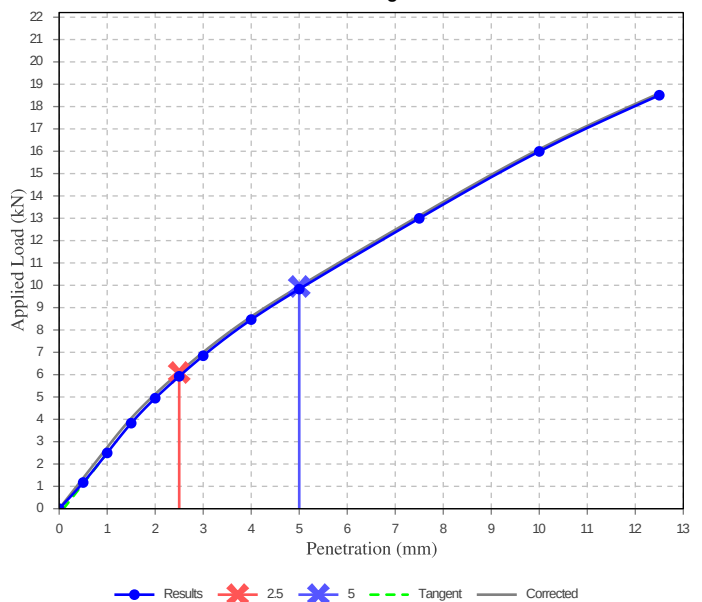
California Bearing Ratio (AS 1289 6.1.1 & 2.1.1)		Min	Max
CBR taken at	5 mm		
CBR %	50		
Method of Compactive Effort	Modified		
Method used to Determine MDD	AS 1289 5.2.1 & 2.1.1		
Method used to Determine Plasticity	Visual Assessment		
Maximum Dry Density (t/m ³)	2.17		
Optimum Moisture Content (%)	7.5		
Laboratory Density Ratio (%)	100.0		
Laboratory Moisture Ratio (%)	102.5		
Dry Density after Soaking (t/m ³)	2.16		
Field Moisture Content (%)	7.5		
Moisture Content at Placement (%)	7.6		
Moisture Content Top 30mm (%)	10.0		
Moisture Content Rest of Sample (%)	8.3		
Mass Surcharge (kg)	4.5		
Soaking Period (days)	4		
Curing Hours	73.5		
Swell (%)	0.0		
Oversize Material (mm)	19		
Oversize Material Included	Excluded		
Oversize Material (%)	0		

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	19		
Plastic Limit (%)	13		
Plasticity Index (%)	6		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	3.0		
Cracking Crumbling Curling	Cracking		

Moisture Content (AS 1289 2.1.1)	
Moisture Content (%)	7.5

California Bearing Ratio



Material Test Report



[Signature]

Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

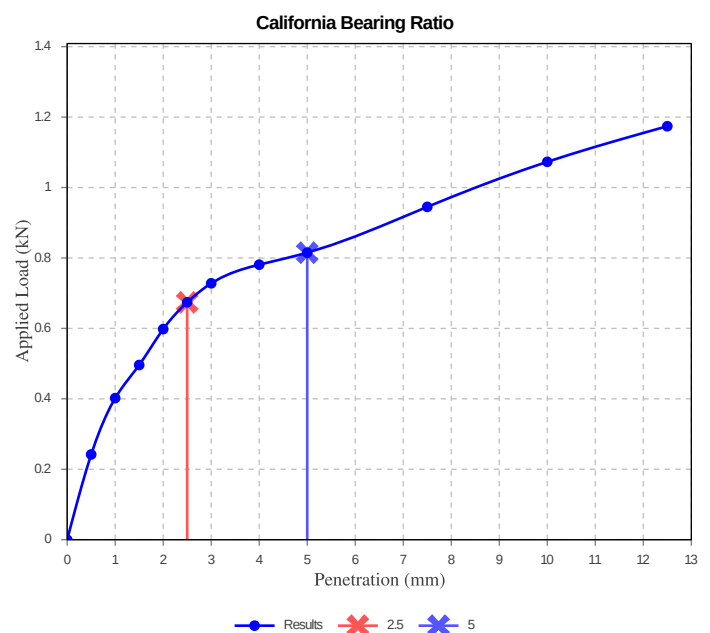
Report Number: 94417.00-1
Issue Number: 1
Date Issued: 12/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3498
Sample Number: GU-3498C
Date Sampled: 28/01/2020
Dates Tested: 28/01/2020 - 07/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Sample Location: Bore 6, Depth: 0.7-0.9m
Material: Sandy Clay

California Bearing Ratio (AS 1289 6.1.1 & 2.1.1)		Min	Max
CBR taken at	2.5 mm		
CBR %	5		
Method of Compactive Effort	Modified		
Method used to Determine MDD	AS 1289 5.2.1 & 2.1.1		
Method used to Determine Plasticity	Visual Assessment		
Maximum Dry Density (t/m ³)	1.67		
Optimum Moisture Content (%)	19.5		
Laboratory Density Ratio (%)	100.5		
Laboratory Moisture Ratio (%)	97.5		
Dry Density after Soaking (t/m ³)	1.63		
Field Moisture Content (%)	8.9		
Moisture Content at Placement (%)	19.1		
Moisture Content Top 30mm (%)	26.2		
Moisture Content Rest of Sample (%)	22.9		
Mass Surcharge (kg)	4.5		
Soaking Period (days)	4		
Curing Hours	71		
Swell (%)	3.0		
Oversize Material (mm)	19		
Oversize Material Included	Excluded		
Oversize Material (%)	0		

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	32		
Plastic Limit (%)	15		
Plasticity Index (%)	17		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	8.5		
Cracking Crumbling Curling	Curling		

Moisture Content (AS 1289 2.1.1)		Min	Max
Moisture Content (%)			8.9



Material Test Report



[Signature]

Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

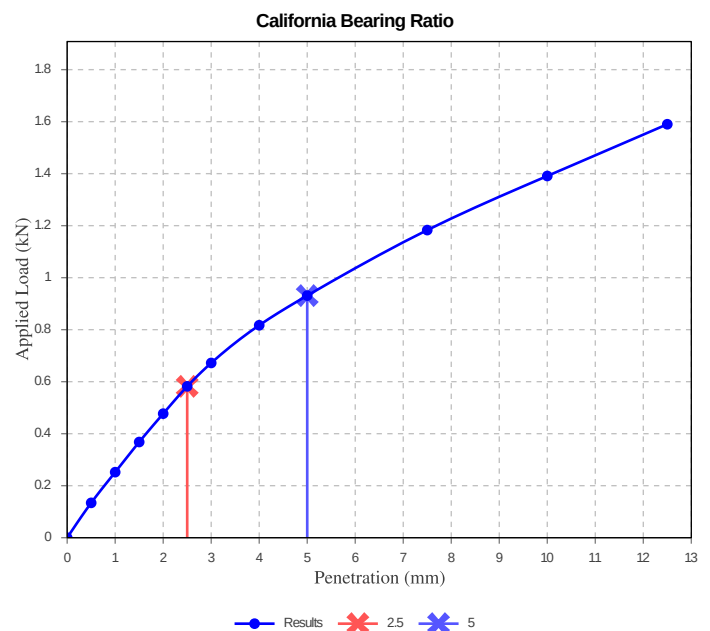
Report Number: 94417.00-1
Issue Number: 1
Date Issued: 12/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3498
Sample Number: GU-3498D
Date Sampled: 28/01/2020
Dates Tested: 28/01/2020 - 07/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Sample Location: Bore 8, Depth: 0.4-0.6m
Material: Sandy Clay

California Bearing Ratio (AS 1289 6.1.1 & 2.1.1)		Min	Max
CBR taken at	5 mm		
CBR %	4.5		
Method of Compactive Effort	Modified		
Method used to Determine MDD	AS 1289 5.2.1 & 2.1.1		
Method used to Determine Plasticity	Visual Assessment		
Maximum Dry Density (t/m ³)	1.90		
Optimum Moisture Content (%)	13.5		
Laboratory Density Ratio (%)	100.5		
Laboratory Moisture Ratio (%)	97.5		
Dry Density after Soaking (t/m ³)	1.81		
Field Moisture Content (%)	12.9		
Moisture Content at Placement (%)	13.0		
Moisture Content Top 30mm (%)	24.4		
Moisture Content Rest of Sample (%)	17.0		
Mass Surcharge (kg)	4.5		
Soaking Period (days)	4		
Curing Hours	72		
Swell (%)	5.5		
Oversize Material (mm)	19		
Oversize Material Included	Excluded		
Oversize Material (%)	0		

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	31		
Plastic Limit (%)	16		
Plasticity Index (%)	15		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	8.0		
Cracking Crumbling Curling	Cracking & Curling		

Moisture Content (AS 1289 2.1.1)	
Moisture Content (%)	12.9



Material Test Report



Approved Signatory: Brett Rippingale

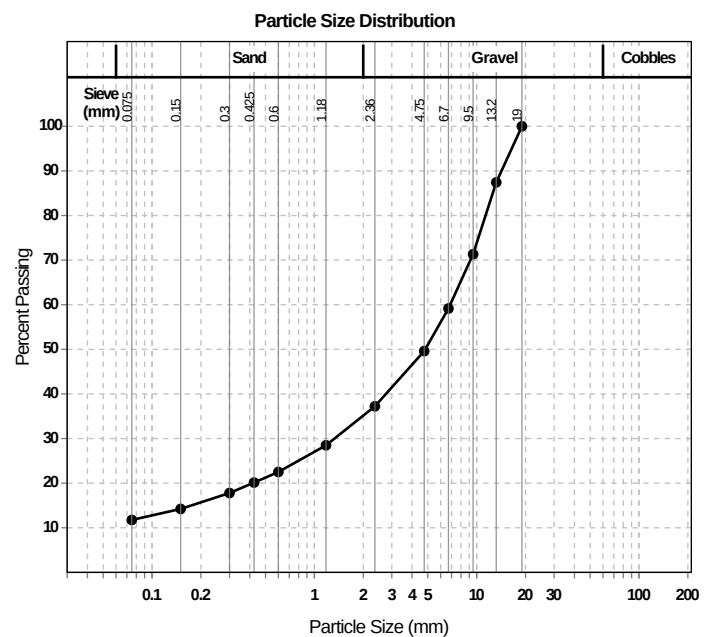
Senior Associate

NATA Accredited Laboratory Number: 828

Report Number: 94417.00-1
Issue Number: 1
Date Issued: 12/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3498
Sample Number: GU-3498E
Date Sampled: 28/01/2020
Dates Tested: 28/01/2020 - 29/01/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Sample Location: Bore 1, Depth: 0.2m
Material: Sandy Gravel

Particle Size Distribution (AS1289 3.6.1)		
Sieve	Passed %	Passing Limits
19 mm	100	
13.2 mm	87	
9.5 mm	71	
6.7 mm	59	
4.75 mm	50	
2.36 mm	37	
1.18 mm	28	
0.6 mm	23	
0.425 mm	20	
0.3 mm	18	
0.15 mm	14	
0.075 mm	12	

Moisture Content (AS 1289 2.1.1)	
Moisture Content (%)	1.6



Material Test Report

Report Number: 94417.00-1
Issue Number: 1
Date Issued: 12/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3498
Sample Number: GU-3498F
Date Sampled: 28/01/2020
Dates Tested: 28/01/2020 - 06/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Sample Location: Bore 2, Depth: 0.9m
Material: Sandy Clay



Douglas Partners Pty Ltd

Goulburn Laboratory

54 Sinclair Street Goulburn NSW 2580

Phone: 02 4822 8395

Email: brett.rippingale@douglaspartners.com.au

Accredited for compliance with ISO/IEC 17025 - Testing



Approved Signatory: Brett Rippingale
Senior Associate

NATA Accredited Laboratory Number: 828

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	39		
Plastic Limit (%)	16		
Plasticity Index (%)	23		
Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	11.0		
Cracking Crumbling Curling	Cracking		
Moisture Content (AS 1289 2.1.1)			
Moisture Content (%)		16.4	

Material Test Report



[Signature]

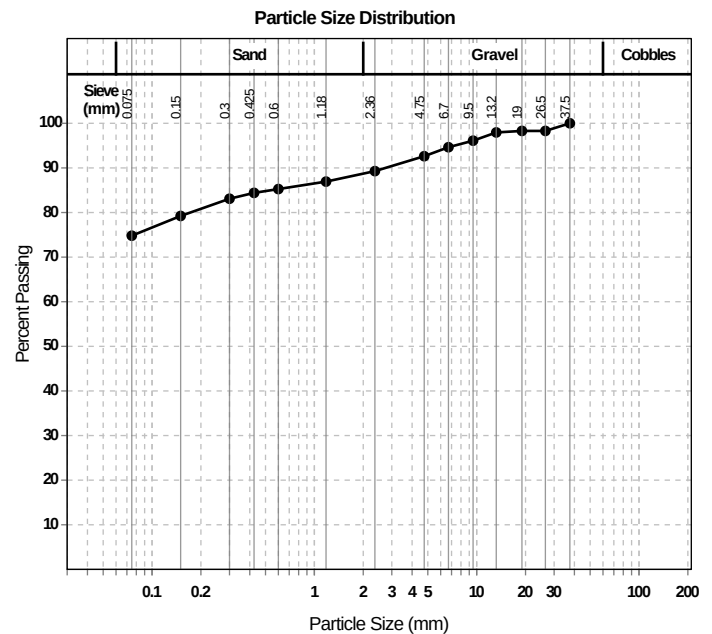
Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Report Number: 94417.00-1
Issue Number: 1
Date Issued: 12/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3498
Sample Number: GU-3498G
Date Sampled: 28/01/2020
Dates Tested: 28/01/2020 - 31/01/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Sample Location: Bore 4, Depth: 0.1m
Material: Sandy Gravelly Clay

Particle Size Distribution (AS1289 3.6.1)		
Sieve	Passed %	Passing Limits
37.5 mm	100	
26.5 mm	98	
19 mm	98	
13.2 mm	98	
9.5 mm	96	
6.7 mm	95	
4.75 mm	93	
2.36 mm	89	
1.18 mm	87	
0.6 mm	85	
0.425 mm	84	
0.3 mm	83	
0.15 mm	79	
0.075 mm	75	
Moisture Content (AS 1289 2.1.1)		
Moisture Content (%)	15.0	



Material Test Report

Report Number: 94417.00-1
Issue Number: 1
Date Issued: 12/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3498
Sample Number: GU-3498H
Date Sampled: 28/01/2020
Dates Tested: 28/01/2020 - 06/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Sample Location: Bore 9, Depth: 0.5m
Material: Sandy Silty Gravel



Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	Not Obtainable		
Plastic Limit (%)	Not Obtainable		
Plasticity Index (%)	Non Plastic		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)			
Cracking Crumbling Curling	None		

Moisture Content (AS 1289 2.1.1)	
Moisture Content (%)	2.2

Material Test Report



[Signature]

Approved Signatory: Brett Rippingale

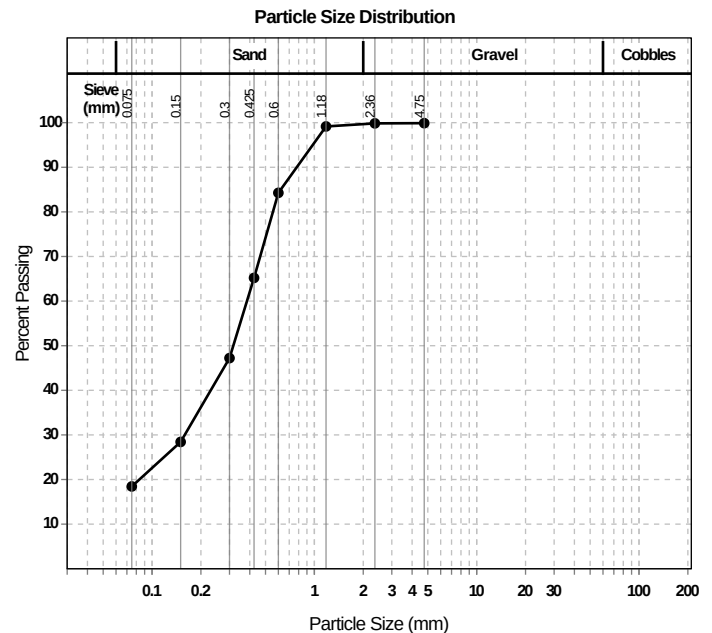
Senior Associate

NATA Accredited Laboratory Number: 828

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582B
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 13/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Remarks: Mix All 3 bags together
Sample Location: BH 22 (0 - 2m)
Material: Sand

Particle Size Distribution (AS1289 3.6.1)		
Sieve	Passed %	Passing Limits
2.36 mm	100	
1.18 mm	99	
0.6 mm	84	
0.425 mm	65	
0.3 mm	47	
0.15 mm	28	
0.075 mm	18	

Moisture Content (AS 1289 2.1.1)	
Moisture Content (%)	2.7



Material Test Report



[Signature]

Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582C
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 19/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: BH 16 (2.4m)
Material: Sandy clay

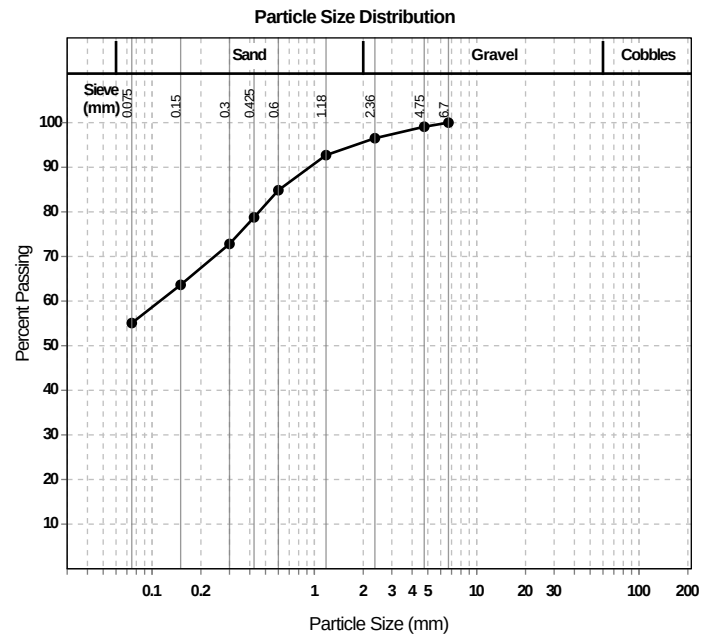
Particle Size Distribution (AS1289 3.6.1)		
Sieve	Passed %	Passing Limits
6.7 mm	100	
4.75 mm	99	
2.36 mm	97	
1.18 mm	93	
0.6 mm	85	
0.425 mm	79	
0.3 mm	73	
0.15 mm	64	
0.075 mm	55	

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Air Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	29		
Plastic Limit (%)	12		
Plasticity Index (%)	17		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	8.0		
Cracking Crumbling Curling	Cracking		

Emerson Class Number of a Soil (AS 1289 3.8.1)		Min	Max
Emerson Class	3		
Soil Description	Red Brown Sandy Clay		
Nature of Water	Distilled		
Temperature of Water (°C)	21		

Moisture Content (AS 1289 2.1.1)	
Moisture Content (%)	9.0



Material Test Report



[Signature]

Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582D
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 18/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: BH 19 (5.5 - 5.8m)
Material: Gravelly Clay

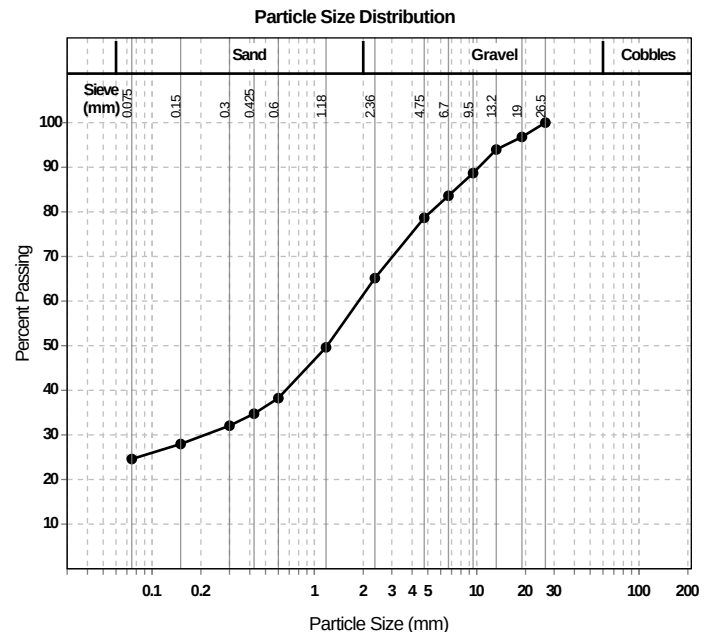
Particle Size Distribution (AS1289 3.6.1)		
Sieve	Passed %	Passing Limits
26.5 mm	100	
19 mm	97	
13.2 mm	94	
9.5 mm	89	
6.7 mm	84	
4.75 mm	79	
2.36 mm	65	
1.18 mm	50	
0.6 mm	38	
0.425 mm	35	
0.3 mm	32	
0.15 mm	28	
0.075 mm	25	

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	37		
Plastic Limit (%)	16		
Plasticity Index (%)	21		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	10.5		
Cracking Crumbling Curling	None		

Emerson Class Number of a Soil (AS 1289 3.8.1)		Min	Max
Emerson Class	3		
Soil Description	Light Brown Gravelly Clay		
Nature of Water	Distilled		
Temperature of Water (°C)	21		

Moisture Content (AS 1289 2.1.1)	
Moisture Content (%)	8.1



Material Test Report



[Signature]

Approved Signatory: Brett Rippingale

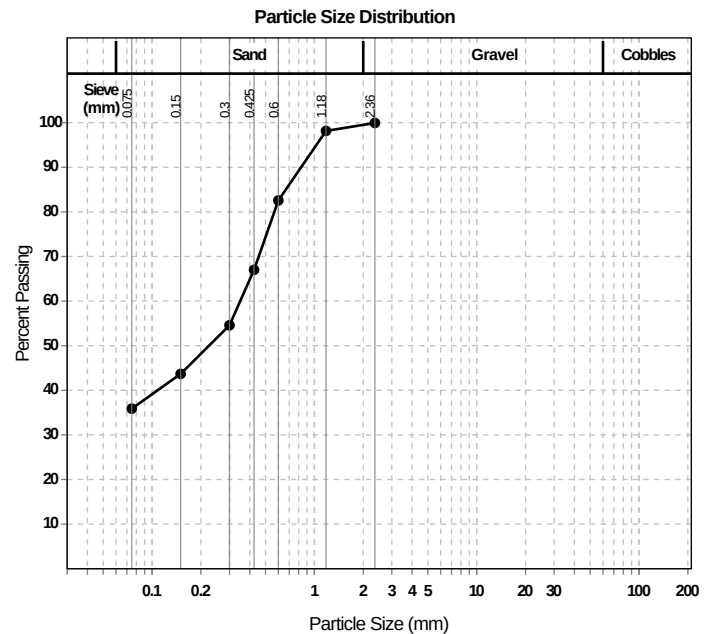
Senior Associate

NATA Accredited Laboratory Number: 828

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582E
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 13/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: BH 18 (1 - 1.45m)
Material: Sand

Particle Size Distribution (AS1289 3.6.1)		
Sieve	Passed %	Passing Limits
2.36 mm	100	
1.18 mm	98	
0.6 mm	83	
0.425 mm	67	
0.3 mm	55	
0.15 mm	44	
0.075 mm	36	

Moisture Content (AS 1289 2.1.1)	
Moisture Content (%)	6.8



Material Test Report



[Signature]

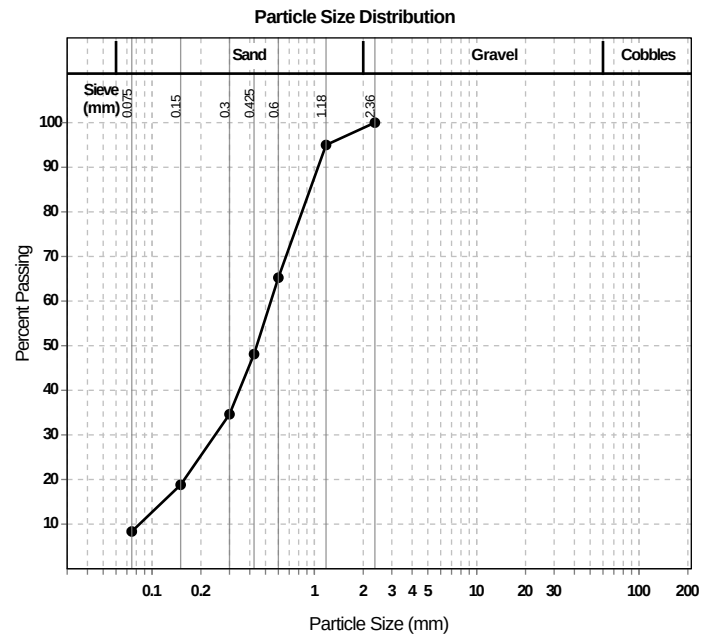
Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582F
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 14/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: BH 21 (1 - 1.45 m)
Material: Clayey Sand

Particle Size Distribution (AS1289 3.6.1)		
Sieve	Passed %	Passing Limits
2.36 mm	100	
1.18 mm	95	
0.6 mm	65	
0.425 mm	48	
0.3 mm	35	
0.15 mm	19	
0.075 mm	8	
Moisture Content (AS 1289 2.1.1)		
Moisture Content (%)	7.7	



Material Test Report



[Signature]

Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582G
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 19/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: BH 21 (5.5 - 5.95m)
Material: Sandy Clay

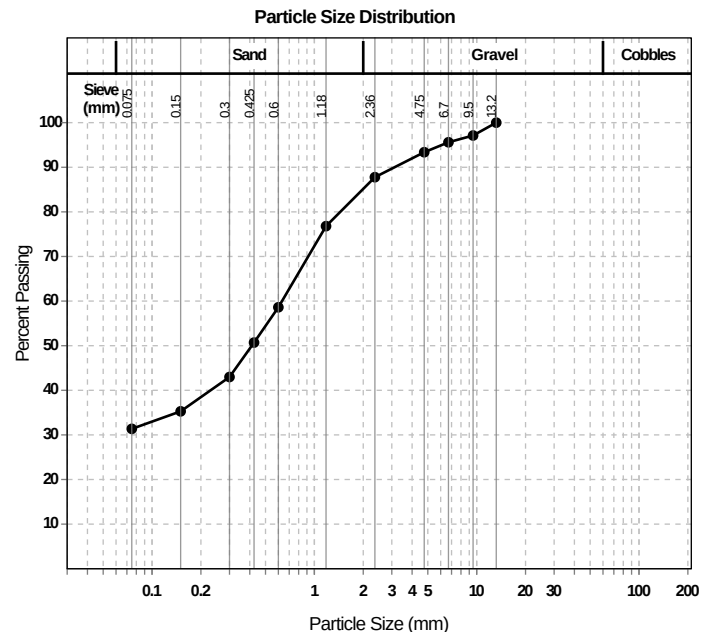
Particle Size Distribution (AS1289 3.6.1)		
Sieve	Passed %	Passing Limits
13.2 mm	100	
9.5 mm	97	
6.7 mm	96	
4.75 mm	93	
2.36 mm	88	
1.18 mm	77	
0.6 mm	59	
0.425 mm	51	
0.3 mm	43	
0.15 mm	35	
0.075 mm	31	

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Air Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	36		
Plastic Limit (%)	13		
Plasticity Index (%)	23		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	10.0		
Cracking Crumbling Curling	None		

Emerson Class Number of a Soil (AS 1289 3.8.1)		Min	Max
Emerson Class	3		
Soil Description	Light Brown Sandy Clay		
Nature of Water	Distilled		
Temperature of Water (°C)	21		

Moisture Content (AS 1289 2.1.1)	
Moisture Content (%)	9.8



Material Test Report



[Signature]

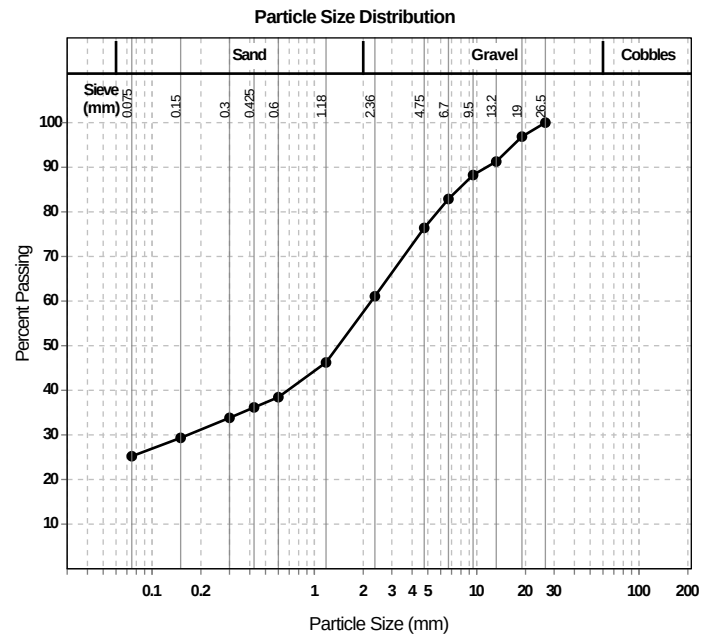
Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582H
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 13/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: BH 36 (1.5 - 1.8m)
Material: Clayey Gravel

Particle Size Distribution (AS1289 3.6.1)		
Sieve	Passed %	Passing Limits
26.5 mm	100	
19 mm	97	
13.2 mm	91	
9.5 mm	88	
6.7 mm	83	
4.75 mm	76	
2.36 mm	61	
1.18 mm	46	
0.6 mm	38	
0.425 mm	36	
0.3 mm	34	
0.15 mm	29	
0.075 mm	25	
Moisture Content (AS 1289 2.1.1)		
Moisture Content (%)	4.1	



Material Test Report

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582I
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 18/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: BH 40 (1 - 1.45m)
Material: Fill/Sandy silt



Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	24		
Plastic Limit (%)	18		
Plasticity Index (%)	6		

Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	4.0		
Cracking Crumbling Curling	None		

Emerson Class Number of a Soil (AS 1289 3.8.1)		Min	Max
Emerson Class	5		
Soil Description	Light Brown Sandy Silt		
Nature of Water	Distilled		
Temperature of Water (°C)	21		

Moisture Content (AS 1289 2.1.1)	
Moisture Content (%)	2.3

Material Test Report

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582J
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 19/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: BH 18 (4.0 - 4.45m)
Material: Silty clay



Approved Signatory: Brett Rippingale
Senior Associate

NATA Accredited Laboratory Number: 828

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	25		
Plastic Limit (%)	10		
Plasticity Index (%)	15		
Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	6.0		
Cracking Crumbling Curling	None		
Moisture Content (AS 1289 2.1.1)			
Moisture Content (%)		10.2	

Material Test Report

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582K
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 18/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: BH 24 (2.5 - 2.95)
Material: Sandy Silty Clay



Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	33		
Plastic Limit (%)	11		
Plasticity Index (%)	22		
Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	8.0		
Cracking Crumbling Curling	Cracking		
Moisture Content (AS 1289 2.1.1)			
Moisture Content (%)		7.2	

Material Test Report

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582L
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 18/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: BH 31 (1 - 1.45m)
Material: Silty Clay



Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	43		
Plastic Limit (%)	14		
Plasticity Index (%)	29		
Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	8.0		
Cracking Crumbling Curling	Cracking & Curling		
Moisture Content (AS 1289 2.1.1)			
Moisture Content (%)		6.1	

Material Test Report

Report Number: 94417.00-3
Issue Number: 1
Date Issued: 24/02/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Contact: Daniel Ballinger
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 3582
Sample Number: GU-3582M
Date Sampled: 21/01/2020
Dates Tested: 11/02/2020 - 18/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: BH 33 (2.5 - 2.95m)
Material: Sandy Silty Clay



Approved Signatory: Brett Rippingale

Senior Associate

NATA Accredited Laboratory Number: 828

Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	23		
Plastic Limit (%)	14		
Plasticity Index (%)	9		
Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	6.0		
Cracking Crumbling Curling	None		
Moisture Content (AS 1289 2.1.1)			
Moisture Content (%)		10.0	

Material Test Report

Report Number: 94417.00-4
Issue Number: 1
Date Issued: 01/03/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 4562
Sample Number: NC-4562A
Date Sampled: 21/01/2020
Dates Tested: 17/02/2020 - 25/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: 29, Depth: 2.5-2.95m



Douglas Partners

Geotechnics | Environment | Groundwater

Douglas Partners Pty Ltd

Newcastle Laboratory

15 Callistemon Close Warabrook Newcastle NSW 2310

Phone: (02) 4960 9600

Fax: (02) 4960 9601

Email: Peter.Gorseski@douglaspartners.com.au

Accredited for compliance with ISO/IEC 17025 - Testing



Approved Signatory: Peter Gorseski
Laboratory Manager
NATA Accredited Laboratory Number: 828

Moisture Content (AS 1289 2.1.1)			
Moisture Content (%)	9.5		
Emerson Class Number of a Soil (AS 1289 3.8.1)			
Emerson Class	2	Min	Max
Soil Description	Clay		
Nature of Water	Distilled		
Temperature of Water (°C)	25		
Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)			
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	37		
Plastic Limit (%)	17		
Plasticity Index (%)	20		
Linear Shrinkage (AS1289 3.4.1)			
Linear Shrinkage (%)	11.0	Min	Max
Cracking Crumbling Curling	Cracking & Curling		

Material Test Report

Report Number: 94417.00-4
Issue Number: 1
Date Issued: 01/03/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 4562
Sample Number: NC-4562B
Date Sampled: 21/01/2020
Dates Tested: 17/02/2020 - 25/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: 20, Depth: 3.9-4.1m



Douglas Partners

Geotechnics | Environment | Groundwater

Douglas Partners Pty Ltd

Newcastle Laboratory

15 Callistemon Close Warabrook Newcastle NSW 2310

Phone: (02) 4960 9600

Fax: (02) 4960 9601

Email: Peter.Gorseski@douglaspartners.com.au

Accredited for compliance with ISO/IEC 17025 - Testing



Approved Signatory: Peter Gorseski
Laboratory Manager
NATA Accredited Laboratory Number: 828

Moisture Content (AS 1289 2.1.1)			
Moisture Content (%)	9.9		
Emerson Class Number of a Soil (AS 1289 3.8.1)			
Emerson Class	2	Min	Max
Soil Description	Clay		
Nature of Water	Distilled		
Temperature of Water (°C)	25		
Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)			
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	29		
Plastic Limit (%)	13		
Plasticity Index (%)	16		
Linear Shrinkage (AS1289 3.4.1)			
Linear Shrinkage (%)	9.0		
Cracking Crumbling Curling	Cracking		

Material Test Report

Report Number: 94417.00-4
Issue Number: 1
Date Issued: 01/03/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 4562
Sample Number: NC-4562C
Date Sampled: 21/01/2020
Dates Tested: 17/02/2020 - 24/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: 19, Depth: 2.5m
Material: Silty Clay



Approved Signatory: Peter Gorseski
Laboratory Manager
NATA Accredited Laboratory Number: 828

Moisture Content (AS 1289 2.1.1)			
Moisture Content (%)	11.6		
Emerson Class Number of a Soil (AS 1289 3.8.1)		Min	Max
Emerson Class	4 *		
Soil Description	Clay		
Nature of Water	Distilled		
Temperature of Water (°C)	25		
* Mineral Present	Carbonate		
Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	26		
Plastic Limit (%)	9		
Plasticity Index (%)	17		
Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	9.5		
Cracking Crumbling Curling	Cracking		

Material Test Report



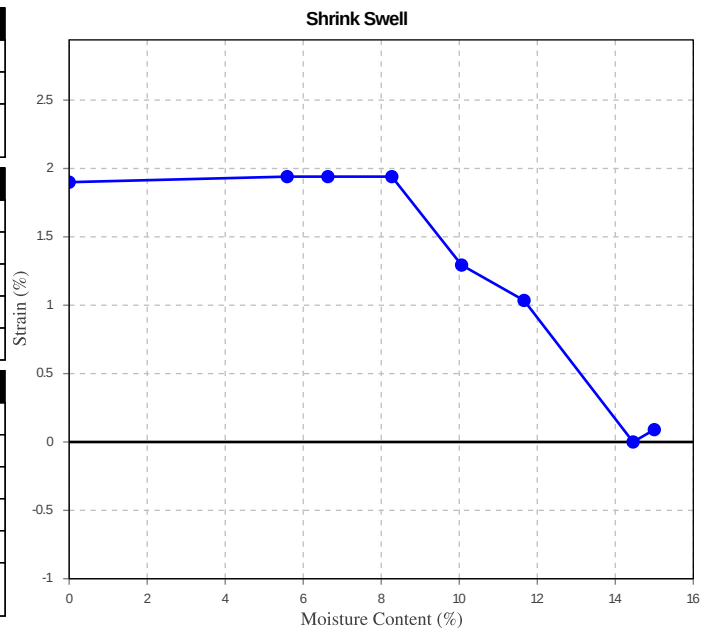
Approved Signatory: Peter Gorseski
Laboratory Manager
NATA Accredited Laboratory Number: 828

Report Number: 94417.00-4
Issue Number: 1
Date Issued: 01/03/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 4562
Sample Number: NC-4562C
Date Sampled: 21/01/2020
Dates Tested: 17/02/2020 - 17/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: 19, Depth: 2.5m
Material: Silty Clay

Shrink Swell Index (AS 1289 7.1.1 & 2.1.1)	
Iss (%)	1.1
Visual Description	Silty Clay
* Shrink Swell Index (Iss) reported as the percentage vertical strain per pF change in suction.	

Core Shrinkage Test	
Shrinkage Strain - Oven Dried (%)	1.9
Estimated % by volume of significant inert inclusions	0
Cracking	Uncracked
Crumbling	No
Moisture Content (%)	14.5

Swell Test	
Initial Pocket Penetrometer (kPa)	>600
Final Pocket Penetrometer (kPa)	450
Initial Moisture Content (%)	14.6
Final Moisture Content (%)	15.0
Swell (%)	-0.1
* NATA Accreditation does not cover the performance of pocket penetrometer readings.	



Material Test Report

Report Number: 94417.00-4
Issue Number: 1
Date Issued: 01/03/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 4562
Sample Number: NC-4562D
Date Sampled: 21/01/2020
Dates Tested: 17/02/2020 - 21/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: 32, Depth: 1.0-1.1m



Douglas Partners Pty Ltd
Newcastle Laboratory
15 Callistemon Close Warabrook Newcastle NSW 2310
Phone: (02) 4960 9600
Fax: (02) 4960 9601
Email: Peter.Gorseski@douglaspartners.com.au



Accredited for compliance with ISO/IEC 17025 - Testing


Approved Signatory: Peter Gorseski
Laboratory Manager
NATA Accredited Laboratory Number: 828

Moisture Content (AS 1289 2.1.1)			
Moisture Content (%)		9.3	
Emerson Class Number of a Soil (AS 1289 3.8.1)		Min	Max
Emerson Class	2		
Soil Description	Silty Clay some Sand		
Nature of Water	Distilled		
Temperature of Water (°C)	25		
Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)		Min	Max
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	38		
Plastic Limit (%)	9		
Plasticity Index (%)	29		
Linear Shrinkage (AS1289 3.4.1)		Min	Max
Linear Shrinkage (%)	12.5		
Cracking Crumbling Curling	Cracking		

Material Test Report

Report Number: 94417.00-4
Issue Number: 1
Date Issued: 01/03/2020
Client: Hunter H2O Holdings Pty Ltd
 PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 4562
Sample Number: NC-4562E
Date Sampled: 21/01/2020
Dates Tested: 17/02/2020 - 25/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: 17, Depth: 1.5-1.95m




Approved Signatory: Peter Gorseski
 Laboratory Manager
 NATA Accredited Laboratory Number: 828

Moisture Content (AS 1289 2.1.1)			
Moisture Content (%)	11.9		
Emerson Class Number of a Soil (AS 1289 3.8.1)			
Emerson Class	4 *	Min	Max
Soil Description	Sandy CLAY		
Nature of Water	Distilled		
Temperature of Water (°C)	24		
* Mineral Present	Carbonate		
Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)			
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	17		
Plastic Limit (%)	14		
Plasticity Index (%)	3		
Linear Shrinkage (AS1289 3.4.1)			
Linear Shrinkage (%)	2.5		
Cracking Crumbling Curling	Cracking		

Material Test Report

Report Number: 94417.00-4
Issue Number: 1
Date Issued: 01/03/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 4562
Sample Number: NC-4562F
Date Sampled: 21/01/2020
Dates Tested: 17/02/2020 - 25/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: 15, Depth: 4.0-4.3m
Material: Clay



Douglas Partners Pty Ltd
Newcastle Laboratory
15 Callistemon Close Warabrook Newcastle NSW 2310
Phone: (02) 4960 9600
Fax: (02) 4960 9601
Email: Peter.Gorseski@douglaspartners.com.au



Accredited for compliance with ISO/IEC 17025 - Testing

Approved Signatory: Peter Gorseski
Laboratory Manager
NATA Accredited Laboratory Number: 828

Moisture Content (AS 1289 2.1.1)			
Moisture Content (%)	11.4		
Emerson Class Number of a Soil (AS 1289 3.8.1)			
Emerson Class	3	Min	Max
Soil Description	CLAY		
Nature of Water	Distilled		
Temperature of Water (°C)	25		
Atterberg Limit (AS1289 3.1.2 & 3.2.1 & 3.3.1)			
Sample History	Oven Dried		
Preparation Method	Dry Sieve		
Liquid Limit (%)	29		
Plastic Limit (%)	12		
Plasticity Index (%)	17		
Linear Shrinkage (AS1289 3.4.1)			
Linear Shrinkage (%)	10.0	Min	Max
Cracking Crumbling Curling	Cracking		

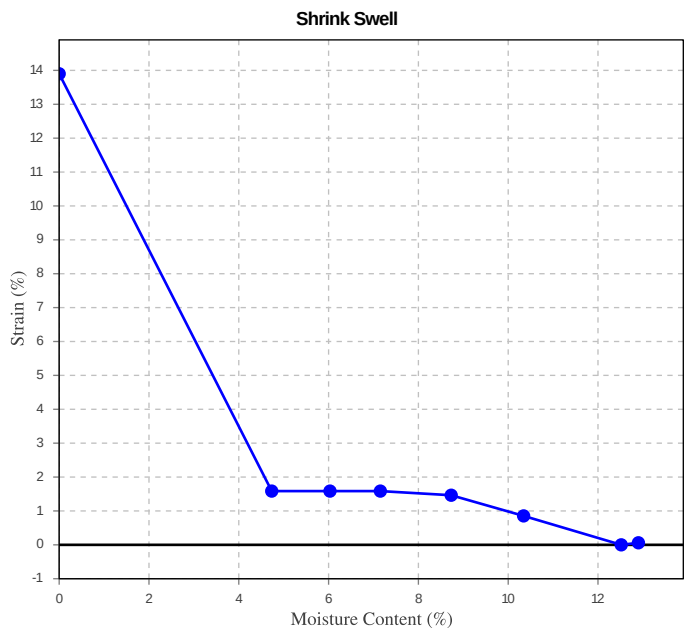
Material Test Report



Approved Signatory: Peter Gorseski
Laboratory Manager
NATA Accredited Laboratory Number: 828

Report Number: 94417.00-4
Issue Number: 1
Date Issued: 01/03/2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Project
Project Location: Mountain Road, Jerrabomberra
Work Request: 4562
Sample Number: NC-4562F
Date Sampled: 21/01/2020
Dates Tested: 17/02/2020 - 17/02/2020
Sampling Method: Sampled by Engineering Department
The results apply to the sample as received
Preparation Method: AS 1289.1.1 - Sampling and preparation of soils
Sample Location: 15, Depth: 4.0-4.3m
Material: Clay

Shrink Swell Index (AS 1289 7.1.1 & 2.1.1)	
Iss (%)	7.7
Visual Description	Clay
* Shrink Swell Index (Iss) reported as the percentage vertical strain per pF change in suction.	
Core Shrinkage Test	
Shrinkage Strain - Oven Dried (%)	13.9
Estimated % by volume of significant inert inclusions	0
Cracking	Moderately Cracked
Crumbling	No
Moisture Content (%)	12.5
Swell Test	
Initial Pocket Penetrometer (kPa)	600
Final Pocket Penetrometer (kPa)	600
Initial Moisture Content (%)	12.6
Final Moisture Content (%)	12.9
Swell (%)	-0.1
* NATA Accreditation does not cover the performance of pocket penetrometer readings.	



Results of Pinhole Dispersion Tests

Client: Hunter H20 Holdings Pty Ltd		Project No: 94417.00	
Project: Queanbeyan Sewage Treatment Plant Upgrade		Report No: 94417.00_4	
Location: Mountain Road, Jerrabomberra		Report Date: 28.02.2020	
		Date Sampled: 21.01.2020	
		Date of Test: 25.02.2020	
		Page: 1 of 1	

Test Location	Depth (m)	FMC %	Description	MC of Sample before Testing (%)	Density of Sample before Testing (t/m ³)	Pinhole Dispersion Classification
Bore 29	2.5-2.95	-	Clay	9.5		D2
Bore 19	2.5	-	Clay	11.6		PD2
Bore 32	1.0-1.1	-	Silty Clay with sand	9.3		D2
Bore 17	1.5-1.95	-	Sandy Clay	11.9		PD2
Bore 15	4.0-4.3	-	Clay	11.4		D2

Dispersion Classification:

D1 Highly dispersive
 D2 Dispersive
 PD1 Potentially dispersive

PD2 Potentially dispersive (intermediate)
 ND1 Non-dispersive
 ND2 Completely erosion resistant

Legend

FMC = Field Moisture Content

MC = Moisture Content

Test Method(s): AS 1289.3.8.3

Sampling Method(s): Sampled by Client

Remarks:



NATA Accredited Laboratory Number: 828

The results of the tests, calibrations and/or measurements included in this document are traceable to Australian/national standards. Accredited for compliance with ISO/IEC 17025

Tested: MB
 Checked: JH

Peter Gorseski
 Laboratory Manager

Uniaxial Compressive Strength

Report Number: 94417.00_2
Issue Number: 1
Date Issued: 24.02.2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Pr
Project Location: Jerrabomberra
Work Request: 4573
Date Sampled: 21.01.2020
Sampling Method: Sampled by Others

Douglas Partners Pty Ltd
Newcastle Laboratory
15 Callistemon Close Warabrook Newcastle NSW 2310
Phone: (02) 4960 9600
Email: peter.gorseski@douglaspartners.com.au

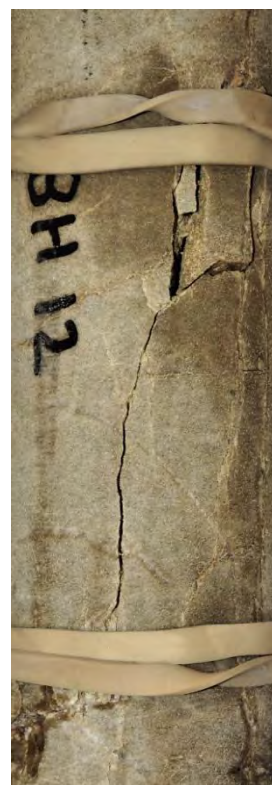


Accredited for Compliance with ISO/IEC 17025 - Testing

Approved Signatory: **Peter Gorseski**
NATA Accredited Laboratory Number: 828

Uniaxial Compressive Strength of Rock Core AS 4133.4.2, AS 4133.1.1.1

Sample Number	NC-4573A
Sample Location	Bore 12
Depth (m)	8.1 - 8.3
Rock Description	
Storage History and Environment	Tested as Received
Orientation to Bedding	-
Compression Machine	Automax Multitest
Date of Testing	20.02.2020
Duration of Test (seconds)	364
Average Diameter (mm)	51.9
Average Height (mm)	141
Height to Diameter Ratio	2.7 : 1
Moisture Content (%)	0.7
Wet Mass / Unit Volume (t/m ³)	2.61
Dry Mass / Unit Volume (t/m ³)	2.60
Uniaxial Compressive Strength (MPa)	122.5
Comments	



Uniaxial Compressive Strength

Report Number: 94417.00_2
Issue Number: 1
Date Issued: 24.02.2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Pr
Project Location: Jerrabomberra
Work Request: 4573
Date Sampled: 21.01.2020
Sampling Method: Sampled by Others

Douglas Partners Pty Ltd
Newcastle Laboratory
15 Callistemon Close Warabrook Newcastle NSW 2310
Phone: (02) 4960 9600
Email: peter.gorseski@douglaspartners.com.au



Accredited for Compliance with ISO/IEC 17025 - Testing

Approved Signatory:

Peter Gorseski

NATA Accredited Laboratory Number: 828

Uniaxial Compressive Strength of Rock Core AS 4133.4.2, AS 4133.1.1.1

Sample Number	NC-4573B
Sample Location	Bore 19
Depth (m)	10.6 - 11.0
Rock Description	
Storage History and Environment	Tested as Received
Orientation to Bedding	-
Compression Machine	Automax Multitest
Date of Testing	20.02.2020
Duration of Test (seconds)	5
Average Diameter (mm)	51.6
Average Height (mm)	139
Height to Diameter Ratio	2.7 : 1
Moisture Content (%)	8.5
Wet Mass / Unit Volume (t/m ³)	2.40
Dry Mass / Unit Volume (t/m ³)	2.22
Uniaxial Compressive Strength (MPa)	5.5
Comments	



Uniaxial Compressive Strength

Report Number: 94417.00_2
Issue Number: 1
Date Issued: 24.02.2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Pr
Project Location: Jerrabomberra
Work Request: 4573
Date Sampled: 21.01.2020
Sampling Method: Sampled by Others

Douglas Partners Pty Ltd
Newcastle Laboratory
15 Callistemon Close Warabrook Newcastle NSW 2310
Phone: (02) 4960 9600
Email: peter.gorseski@douglaspartners.com.au



Accredited for Compliance with ISO/IEC 17025 - Testing

Approved Signatory:

Peter Gorseski

NATA Accredited Laboratory Number: 828

Uniaxial Compressive Strength of Rock Core AS 4133.4.2, AS 4133.1.1.1

Sample Number	NC-4573C
Sample Location	Bore 25
Depth (m)	9.0 - 9.25
Rock Description	
Storage History and Environment	Tested as Received
Orientation to Bedding	-
Compression Machine	Automax Multitest
Date of Testing	20.02.2020
Duration of Test (seconds)	2
Average Diameter (mm)	51.6
Average Height (mm)	146
Height to Diameter Ratio	2.8 : 1
Moisture Content (%)	2.2
Wet Mass / Unit Volume (t/m^3)	2.56
Dry Mass / Unit Volume (t/m^3)	2.50
Uniaxial Compressive Strength (MPa)	4.6
Comments	



Uniaxial Compressive Strength

Report Number: 94417.00_2
Issue Number: 1
Date Issued: 24.02.2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Pr
Project Location: Jerrabomberra
Work Request: 4573
Date Sampled: 21.01.2020
Sampling Method: Sampled by Others

Douglas Partners Pty Ltd
Newcastle Laboratory
15 Callistemon Close Warabrook Newcastle NSW 2310
Phone: (02) 4960 9600
Email: peter.gorseski@douglaspartners.com.au



Accredited for Compliance with ISO/IEC 17025 - Testing

Approved Signatory:

Peter Gorseski

NATA Accredited Laboratory Number: 828

Uniaxial Compressive Strength of Rock Core AS 4133.4.2, AS 4133.1.1.1

Sample Number	NC-4573D
Sample Location	Bore 30
Depth (m)	4.6 - 4.8
Rock Description	
Storage History and Environment	Tested as Received
Orientation to Bedding	-
Compression Machine	Automax Multitest
Date of Testing	20.02.2020
Duration of Test (seconds)	5
Average Diameter (mm)	51.6
Average Height (mm)	133
Height to Diameter Ratio	2.6 : 1
Moisture Content (%)	3.4
Wet Mass / Unit Volume (t/m ³)	2.59
Dry Mass / Unit Volume (t/m ³)	2.50
Uniaxial Compressive Strength (MPa)	3.1
Comments	



Uniaxial Compressive Strength

Report Number: 94417.00_2
Issue Number: 1
Date Issued: 24.02.2020
Client: Hunter H2O Holdings Pty Ltd
PO Box 5007, Hunter Region Mail Centre NSW 2310
Project Number: 94417.00
Project Name: Queanbeyan Sewage Treatment Plant Upgrade Pr
Project Location: Jerrabomberra
Work Request: 4573
Date Sampled: 21.01.2020
Sampling Method: Sampled by Others

Douglas Partners Pty Ltd
Newcastle Laboratory
15 Callistemon Close Warabrook Newcastle NSW 2310
Phone: (02) 4960 9600
Email: peter.gorseski@douglaspartners.com.au



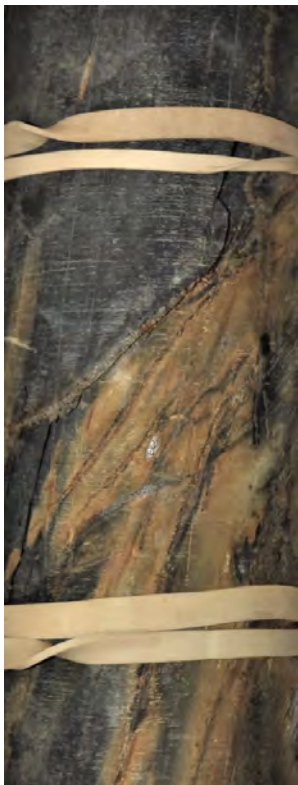
Accredited for Compliance with ISO/IEC 17025 - Testing

Approved Signatory:

Peter Gorseski

NATA Accredited Laboratory Number: 828

Uniaxial Compressive Strength of Rock Core AS 4133.4.2, AS 4133.1.1.1

Sample Number	NC-4573E	
Sample Location	Bore 28	
Depth (m)	10.3 - 10.7	
Rock Description		
Storage History and Environment	Tested as Received	
Orientation to Bedding	-	
Compression Machine	Automax Multitest	
Date of Testing	20.02.2020	
Duration of Test (seconds)	23	
Average Diameter (mm)	51.7	
Average Height (mm)	135	
Height to Diameter Ratio	2.6 : 1	
Moisture Content (%)	5.4	
Wet Mass / Unit Volume (t/m ³)	2.51	
Dry Mass / Unit Volume (t/m ³)	2.38	
Uniaxial Compressive Strength (MPa)	3.9	
Comments		

CERTIFICATE OF ANALYSIS 236541

Client Details

Client	Douglas Partners Canberra
Attention	Shannon Goodsell
Address	PO Box 1487, Fyshwick, ACT, 2609

Sample Details

Your Reference	<u>94417.00, QSTP</u>
Number of Samples	10 soil
Date samples received	12/02/2020
Date completed instructions received	12/02/2020

Analysis Details

Please refer to the following pages for results, methodology summary and quality control data.
Samples were analysed as received from the client. Results relate specifically to the samples as received.
Results are reported on a dry weight basis for solids and on an as received basis for other matrices.
Please refer to the last page of this report for any comments relating to the results.

Report Details

Date results requested by	19/02/2020
Date of Issue	18/02/2020
NATA Accreditation Number 2901. This document shall not be reproduced except in full.	
Accredited for compliance with ISO/IEC 17025 - Testing. Tests not covered by NATA are denoted with *	

Results Approved By

Priya Samarawickrama, Senior Chemist

Authorised By



Nancy Zhang, Laboratory Manager

Soil Aggressivity

Our Reference		236541-1	236541-2	236541-3	236541-4	236541-5
Your Reference	UNITS	BH11	BH10	BH19	BH30	BH21
Date Sampled		10/02/2020	10/02/2020	10/02/2020	10/02/2020	10/02/2020
Type of sample		soil	soil	soil	soil	soil
pH 1:5 soil:water	pH Units	6.9	6.4	6.2	8.6	6.8
Electrical Conductivity 1:5 soil:water	µS/cm	15	61	260	190	180
Chloride, Cl 1:5 soil:water	mg/kg	<10	31	29	55	10
Sulphate, SO4 1:5 soil:water	mg/kg	<10	86	310	120	220

Soil Aggressivity

Our Reference		236541-6	236541-7	236541-8	236541-9	236541-10
Your Reference	UNITS	BH34	BH40	BH27	BH26	BH15
Date Sampled		10/02/2020	10/02/2020	10/02/2020	10/02/2020	10/02/2020
Type of sample		soil	soil	soil	soil	soil
pH 1:5 soil:water	pH Units	6.6	7.6	7.1	7.0	6.8
Electrical Conductivity 1:5 soil:water	µS/cm	210	230	36	43	300
Chloride, Cl 1:5 soil:water	mg/kg	10	<10	20	20	28
Sulphate, SO4 1:5 soil:water	mg/kg	270	120	31	35	140

Method ID	Methodology Summary
Inorg-001	pH - Measured using pH meter and electrode in accordance with APHA latest edition, 4500-H+. Please note that the results for water analyses are indicative only, as analysis outside of the APHA storage times.
Inorg-002	Conductivity and Salinity - measured using a conductivity cell at 25°C in accordance with APHA latest edition 2510 and Rayment & Lyons.
Inorg-081	Anions - a range of Anions are determined by Ion Chromatography, in accordance with APHA latest edition, 4110-B. Waters samples are filtered on receipt prior to analysis. Alternatively determined by colourimetry/turbidity using Discrete Analyser.

QUALITY CONTROL: Soil Aggressivity					Duplicate			Spike Recovery %		
Test Description	Units	PQL	Method	Blank	#	Base	Dup.	RPD	LCS-1	236541-3
pH 1:5 soil:water	pH Units		Inorg-001	[NT]	1	6.9	6.9	0	101	[NT]
Electrical Conductivity 1:5 soil:water	µS/cm	1	Inorg-002	<1	1	15	15	0	105	[NT]
Chloride, Cl 1:5 soil:water	mg/kg	10	Inorg-081	<10	1	<10	<10	0	101	98
Sulphate, SO4 1:5 soil:water	mg/kg	10	Inorg-081	<10	1	<10	<10	0	110	#

Result Definitions

NT	Not tested
NA	Test not required
INS	Insufficient sample for this test
PQL	Practical Quantitation Limit
<	Less than
>	Greater than
RPD	Relative Percent Difference
LCS	Laboratory Control Sample
NS	Not specified
NEPM	National Environmental Protection Measure
NR	Not Reported

Quality Control Definitions

Blank	This is the component of the analytical signal which is not derived from the sample but from reagents, glassware etc, can be determined by processing solvents and reagents in exactly the same manner as for samples.
Duplicate	This is the complete duplicate analysis of a sample from the process batch. If possible, the sample selected should be one where the analyte concentration is easily measurable.
Matrix Spike	A portion of the sample is spiked with a known concentration of target analyte. The purpose of the matrix spike is to monitor the performance of the analytical method used and to determine whether matrix interferences exist.
LCS (Laboratory Control Sample)	This comprises either a standard reference material or a control matrix (such as a blank sand or water) fortified with analytes representative of the analyte class. It is simply a check sample.
Surrogate Spike	Surrogates are known additions to each sample, blank, matrix spike and LCS in a batch, of compounds which are similar to the analyte of interest, however are not expected to be found in real samples.
Australian Drinking Water Guidelines recommend that Thermotolerant Coliform, Faecal Enterococci, & E.Coli levels are less than 1cfu/100mL. The recommended maximums are taken from "Australian Drinking Water Guidelines", published by NHMRC & ARMC 2011.	
The recommended maximums for analytes in urine are taken from "2018 TLVs and BEIs", as published by ACGIH (where available). Limit provided for Nickel is a precautionary guideline as per Position Paper prepared by AIOH Exposure Standards Committee, 2016.	
Guideline limits for Rinse Water Quality reported as per analytical requirements and specifications of AS 4187, Amdt 2 2019, Table 7.2	

Laboratory Acceptance Criteria

Duplicate sample and matrix spike recoveries may not be reported on smaller jobs, however, were analysed at a frequency to meet or exceed NEPM requirements. All samples are tested in batches of 20. The duplicate sample RPD and matrix spike recoveries for the batch were within the laboratory acceptance criteria.

Filters, swabs, wipes, tubes and badges will not have duplicate data as the whole sample is generally extracted during sample extraction.

Spikes for Physical and Aggregate Tests are not applicable.

For VOCs in water samples, three vials are required for duplicate or spike analysis.

Duplicates: >10xPQL - RPD acceptance criteria will vary depending on the analytes and the analytical techniques but is typically in the range 20%-50% – see ELN-P05 QA/QC tables for details; <10xPQL - RPD are higher as the results approach PQL and the estimated measurement uncertainty will statistically increase.

Matrix Spikes, LCS and Surrogate recoveries: Generally 70-130% for inorganics/metals (not SPOCAS); 60-140% for organics/SPOCAS (+/-50% surrogates) and 10-140% for labile SVOCs (including labile surrogates), ultra trace organics and speciated phenols is acceptable.

In circumstances where no duplicate and/or sample spike has been reported at 1 in 10 and/or 1 in 20 samples respectively, the sample volume submitted was insufficient in order to satisfy laboratory QA/QC protocols.

When samples are received where certain analytes are outside of recommended technical holding times (THTs), the analysis has proceeded. Where analytes are on the verge of breaching THTs, every effort will be made to analyse within the THT or as soon as practicable.

Where sampling dates are not provided, Envirolab are not in a position to comment on the validity of the analysis where recommended technical holding times may have been breached.

Measurement Uncertainty estimates are available for most tests upon request.

Analysis of aqueous samples typically involves the extraction/digestion and/or analysis of the liquid phase only (i.e. NOT any settled sediment phase but inclusive of suspended particles if present), unless stipulated on the Envirolab COC and/or by correspondence. Notable exceptions include certain Physical Tests (pH/EC/BOD/COD/Apparent Colour etc.), Solids testing, total recoverable metals and PFAS where solids are included by default.

Samples for Microbiological analysis (not Amoeba forms) received outside of the 2-8°C temperature range do not meet the ideal cooling conditions as stated in AS2031-2012.

Report Comments

Percent recovery is not possible to report due to the high concentration of the element/s in the sample/s. However an acceptable recovery was obtained for the LCS.

Appendix C

Drawing No: SKETCH-C-010_D – Test Location Plan